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Resource allocation strategy for vehicular communication networks based on multi-agent deep reinforcement learning

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ABSTRACT

In complex and high-mobility vehicular communication networks, rapidly changing channel conditions, signal interference, and stringent latency requirements of safety services pose significant challenges to existing wireless resource allocation schemes. We propose a novel resource allocation method named AMADRL. It is based on the multi-agent deep reinforcement learning (MADRL) algorithm and incorporates attention mechanisms (AM). This method first improves the traditional MADRL framework by employing two critic networks to estimate the corresponding global and local reward functions, achieving joint optimization of spectrum and power allocation. This optimization balances the individual interests of agents with the collective benefits, meeting the low-latency communication requirements of vehicle-to-vehicle (V2V) links. And this method effectively reduces the interference to the vehicle-to-infrastructure (V2I) links. Building on this foundation, we further integrate AM into the framework. The AM enables the model to selectively focus on critical information, dynamically adjusting resource allocation strategies. Simulation results demonstrate that, compared with random methods and conventional deep reinforcement learning (DRL) methods, the proposed algorithm exhibits superior convergence speed and stability. It effectively meets the communication requirements of different links and significantly improves spectrum efficiency.

1. Introduction

Vehicle-to-everything (V2X) [1] has become a crucial technology to enhance road safety, entertainment services, and traveling experience [2]. The technology consists of two main communication modes: V2I and V2V. V2I communication connects each vehicle to base stations (BS) and road side units (RSU) and focuses on high-data-rate recreational communication services, consequently demanding high data transfer rates [3]. In contrast, V2V communication provides immediate communication between neighboring vehicles. It focuses on transmitting safety-critical information, such as vehicle heading, to ensure the safety of vehicle movement, which usually has stringent low-latency requirements [4] [5]. However, the differing communication demands of these links present challenges under limited spectrum conditions. Existing studies have proposed enabling the V2V link to reuse the spectrum occupied by the V2I link (occupied orthogonally with constant transmission power) [6]. Thus, we should design resource allocation methods for V2V communication, including the allocation of spectrum sub-bands and transmission power, to meet the communication requirements of different links while enhancing spectrum resource efficiency.

Many studies have used conventional optimization approaches to handle resource allocation concerns in V2X communication. Literature [7] proposes a device-to-device (D2D) based resource management strategy, which represents the spectrum sharing process as a matching problem involving a weighted bipartite graph, and the global optimal answer to this issue is derived by the Hungarian algorithm, which effectively improves the spectrum efficiency. Literature [8] formulates the computational resource allocation problem for edge nodes as an integer linear programming model and proposes a greedy-based heuristic algorithm that guarantees the latency requirements for V2X communications. Simulated annealing methods are also used for power and resource allocation problems in cellular-V2X (C-V2X) communications [9]. Although these methods address the resource allocation problem from various viewpoints, they have certain limitations. First, obtaining accurate channel state information (CSI) is not feasible because of the fast-changing channel conditions in vehicular environments, which incurs high signal overhead [10]. Second, the goal functions of most vehicular communication problems are non-convex and NP-hard, which necessitates the design of iterative methods to gradually compute the response of each channel, resulting in a significant computational bur-

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Table 1
Summary of Shortcomings in Previous Methods.

Literature	Shortcomings
[7–9]	Difficulty in CSI acquisition, complexity of objective functions, and inability to mathematically formalize task requirements.
[14–16]	Single-agent algorithms struggle to coordinate resource allocation among multiple participants.
[19,20,24]	Lacking collaborative optimization, it is difficult to balance individual and team interests. The network structure is simple and unable to analyze the importance of complex network states.
[27–30]	The application of attention mechanisms in spectrum management for vehicular communication is rarely studied.

den and hindering rapid decision-making. Third, with the increasing diversity and complexity of vehicular communication networks, some service requirements, such as the sum-rate and latency constraints in V2X communication, are difficult to accurately express using mathematical methods. Therefore, there is a need for a more intelligent and flexible method to address the resource allocation issue in vehicle networks.

Reinforcement learning (RL) has demonstrated significant advantages in managing resources within vehicular networks in imperfect CSI situations with its powerful learning and adaptive decision-making capabilities [11]. Agents learn by interacting with the environment, tracking the complex and dynamically changing vehicular network environment in real time, and mapping states to behaviors to achieve intelligent decision-making. DRL integrates RL with deep neural networks, capable of modeling and learning complex and high-dimensional state and action spaces with better performance [12] [13]. DRL has been broadly utilized to address resource management issues in vehicle communication networks, providing better services and support to drivers and traffic managers. In the literature [14], the authors constructed a dynamic scheduling framework for cache and computational resources using the deep q-network (DQN) algorithm to enhance the model's performance. The DQN-based distributed framework can also be utilized in resource allocation in different scenarios, effectively solving the problem of delay limitation of V2V links [15]. The authors of the literature [16] developed dual time-scale joint DRL algorithms to boost the efficiency of vehicular network communications. Among them, the classical particle swarm optimization (PSO) method is executed on a larger time scale; in comparison, the DRL algorithm is executed on a smaller time scale, effectively reducing the communication delay. However, as the network size increases, the above single-agent deep reinforcement learning (SADRL) framework has shortcomings, e.g., the SADRL algorithm is limited by local observation information, which makes it challenging to coordinate resource allocation and decision-making among multiple participants efficiently.

The MADRL approach can handle the collaborative and competitive relationships among participants and has become a more attractive solution [17] [18]. Literature [19] introduced a distributed federated MADRL framework which satisfies the transmission reliability of V2V links while enhancing the total V2I link capacity. In [20], the authors applied the multi-agent deep deterministic policy gradient (MADDPG) algorithm [21] within a centralized training and decentralized execution framework. This approach effectively addresses the non-stationarity issue caused by the continuous reshaping of the environment due to interactions among multiple agents in distributed MADRL methods [22] [23]. It also optimizes channel allocation and power control strategies, meeting the heterogeneous quality of service (QoS) requirements in heterogeneous vehicular networks. Literature [24] proposed a resource management model of unmanned-aerial-vehicle (UAV)-assisted vehicular networks using a similar algorithm, which effectively ensures the resources are accessed on demand. However, cooperative multi-agent problems in vehicular communication require greater coordination and collaboration among vehicles, rather than merely maximizing the benefits of individual agents. Most current MADRL methods are trained using a single reward function designed for multi-objective problems, which fails to maximize both individual and team benefits simultaneously or achieve optimal solutions for all objectives. Therefore, improving MADRL methods and designing appropriate reward functions to simultaneously maximize individual and collective benefits has become critically important.

However, the DRL algorithms used in the above literature only construct neural network structures with simple linear layers, making it difficult to analyze the importance of real-time complex vehicular communication network state information. Since the AM enables the model to selectively focus on key information, reduce the processing of redundant information, and improve the model's flexibility and task efficiency, it has been widely applied in complex network environments [25] [26]. Introducing the AM into the DRL model can dynamically modify the attention weights and feature weighting based on environmental changes. Additionally, this mechanism enables dynamic strategy selection by selectively focusing on the features of the state or action to improve the effectiveness of decision-making. For instance, in [27], the authors designed a dual-attention network based on the AM and DRL to solve the flexible job shop scheduling problem (FJSP). In [28], the authors designed an RL network incorporating graph attention networks (GAT) to build and optimize a wind power prediction model. Reference [29] introduced an attention-weighted deep deterministic policy gradient (DDPG) model to solve the complex caching problem in mobile edge computing (MEC) networks, significantly improving cache resource utilization. In the literature [30], the authors incorporated the AM into the MADRL framework to consider the priority of multiple resources in a low-orbit multi-beam satellite-terrestrial network serving vehicles. They realized the on-demand allocation of various resources. However, research on applying AM to wireless spectrum resource management in vehicular communication networks is still limited, which warrants further attention.

Table 1 summarizes the shortcomings of the above methods. The objective of this study is to design a smarter and more efficient resource allocation framework for vehicular networks. Inspired by references [31] and [32], we propose a novel resource allocation method for vehicular networks based on MADRL, named AMADRL. In this method, each V2V link is regarded as an agent that continuously interacts with the environment to acquire information and train itself for resource allocation. This method improves the MADRL network framework, effectively balancing the interests of individual agents and the collective benefits of multiple agents. It enhances spectrum efficiency while meeting the communication requirements of different links. Additionally, the AM is integrated into the framework to improve the model's decision-making efficiency in highly dynamic vehicular environments. This study contributes to providing a new and practical solution to the resource allocation problem in vehicular networks for intelligent transportation systems. The main contributions of this paper are as follows:

1) We model the resource allocation in vehicular communication networks as a multi-agent problem and propose improvements to the MADRL algorithm by training global and local critic networks. The global critic network, similar to the MADDPG algorithm, takes all agents' information as input during the training phase to maintain the collective benefits of multiple agents. In contrast, the local critic networks, resembling the single-agent DDPG algorithm, focus on maintaining the personal interests of individual agents. This improvement addresses the security concerns of traditional MADDPG, where each agent shares in-

formation with others during the training phase. And it also reduces the input size as the number of links increases, thereby enhancing training efficiency.

2) We designed both local rewards and global rewards, which are estimated and maintained by the local and global critic networks, respectively. The local reward function is designed based on the communication requirements of different links, aiming to solve the optimization problem of this study through a multi-objective function. The global reward function is designed based on the total interference between V2V links to promote coordinated operation among multiple agents. This design aims to simultaneously maximize local and global rewards, effectively optimizing resource allocation decisions and balancing competition and cooperation among agents.

3) In this study, the multi-head self-attention mechanism [33] is incorporated into the global critic network of the proposed resource allocation framework. This mechanism helps the model assign different attention weights when processing the input data and improve the model's capability to capture essential input data characteristics. Thus, the model can keenly sense the variations in the vehicle network environment and quickly adjust the resource allocation strategy to guarantee the flexible and efficient functioning of the vehicle communication network.

4) Through extensive simulation experiments, we compared the AMADRL method proposed in this study with baseline methods. The results show that the proposed approach minimizes interference with V2I links and other V2V links while ensuring the latency constraints of V2V link transmissions and demonstrates good convergence performance. Furthermore, we verified the superiority of incorporating the attention mechanism into the model, effectively improving the efficiency and performance of resource allocation in vehicular networks.

The remainder of the paper is arranged as follows. Section 2 provides the system model and formulation of the optimization problem. Section 3 specifies the proposed AMADRL method. Section 4 provides experimental results and a discussion. At last, Section 5 summarizes the article.

2. System model and problem formulation

We describe the architecture for vehicular communication networks studied in this paper first, then model the requirements of different links separately, and finally formulate the research problem of this paper.

2.1. System scenario

This study considers a vehicular communication network, illustrated in Fig. 1. Multiple vehicles travel on the road, and the BS is at an intersection. There are $\mathcal{M} = \{1, 2, ..., M\}$ V2I links and $\mathcal{K} = \{1, 2, ..., K\}$ V2V links. Based on the orthogonal frequency division multiplexing technique, we divide the total authorized bandwidth equally into $\mathcal{F} = \{1, 2, ..., F\}$ sub-bands, each with a bandwidth of W. Assume that each V2I link (considering the uplink) is pre-allocated with orthogonal spectral sub-bands and constant transmission power. Therefore, $|F| = |M|$, i.e., the m-th V2I link occupy the f-th sub-band ($m \in \mathcal{M}$ and $f \in \mathcal{F}$), and V2I links do not experience co-channel interference. Since V2I link communication requires high-rate transmission, while V2V link communication requires strict low latency requirements. To effectively utilize the limited spectrum resources, the V2V links are able to use the same spectrum as that used by V2I links and we need to have an interference signal management design. Therefore, the research is intended to design a resource allocation framework to meet the requirements among different links.

Due to the characteristics of channel fading, different sub-bands are distinct from each other, while within the same sub-band, they are approximately the same. The channel gain of the k-th V2V link on sub-band f within a coherent time slot is $g_k^f = \alpha_k h_k^f$. α_k refers to the large-scale fading (includes path loss and shadowing), which changes very little and

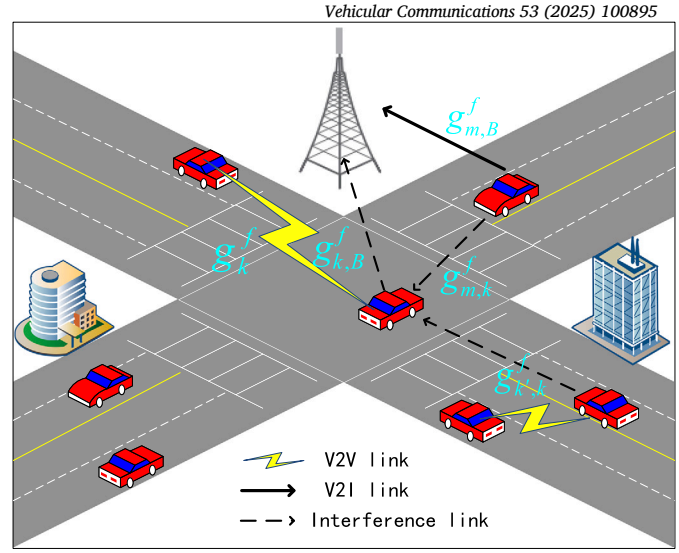

Fig. 1. Vehicle communication network environment.

is usually frequency-independent. h_k^f represents the small-scale fading that depends on frequency, which varies over consecutive time slots and we assume to be distributed as a unit-mean exponential [34]. The channel gain of the m-th V2I link on sub-band f is supposed to be $\tilde{g}_{m,B}^f$. There are three different kinds of interference in the vehicular network considered in this study:

1) The channel interference from the transmitter of the m-th V2I link to the receiver of the k-th V2V link on the f-th sub-band is $\tilde{g}_{m,k}^f$.

2) The channel interference from the transmitter of the k'-th V2V link to the receiver of the k-th V2V link in the f-th sub-band is $g_{k',k}^f$.

3) The channel interference from the k-th V2V link to the BS on the f-th sub-band is $g_{k,B}^f$.

Then, we give the signal-to-interference-plus-noise ratio (SINR) for the m-th V2I link on the f-th sub-band:

$$\Gamma_{\mathrm{m}}^f = \frac{p_m \tilde{g}_{m,B}^f}{\sigma^2 + \sum_{k=1}^K \zeta_k^f p_k^f g_{k,B}^f}, \tag{1}$$

where p_m and p_k^f respectively indicate the transmit power of the m-th V2I transmitter and the k-th V2V transmitter on the f-th sub-band, and σ^2 is the noise power. The $\zeta_k^f \in \{0, 1\}$ represents the sub-band selection index, where $\zeta_k^f = 1$ when the k-th V2V link utilizes the f-th sub-band, and $\zeta_k^f = 0$ otherwise, with the constraint that $\sum_{f=1}^F \zeta_k^f \leq 1$ for all $k \in \mathcal{K}$ and $f \in \mathcal{F}$.

In a similar manner, the SINR corresponding to the k-th V2V link on the f-th sub-band is:

$$\Gamma_k^f = \frac{p_k^f g_k^f}{\sigma^2 + I_{\mathrm{k}}^f}, \tag{2}$$

where $I_{\mathrm{k}}^f = p_m \tilde{g}_{m,k}^f + \sum_{k' \neq k}^K \zeta_{k'}^f p_{k'}^f g_{k',k}^f$ denotes interference from the m-th V2I link on the f-th sub-band and additional V2V links sharing the f-th sub-band. $p_{k'}^f$ represents the transmit power of the k'-th V2V transmitter on the f-th sub-band.

Therefore, the achievable data rates for the m-th V2I and k-th V2V links on the f-th sub-band are:

$$R_{\mathrm{m}}^c = W \log\left(1 + \Gamma_{\mathrm{m}}^f\right), \tag{3}$$

$$R_k^v = W \log\left(1 + \Gamma_k^f\right). \tag{4}$$

2.2. Requirements modelling

Due to the need for high-speed data rates for V2I link transmissions, this utility function can be modeled with respect to the sum rate $\sum_{m=1}^{M} R_{\mathrm{m}}^{c}$ of all V2I link transmissions.

Whereas V2V links have strict requirements on the delay of transmission, we strive to accomplish successful transmission for all V2V users while satisfying the delay constraint. To achieve this aim, we define the ratio of V2V links that successfully complete the transmission task under a given delay constraint:

$$R_{\text{success}} = \frac{K_{\text{satisfied}}}{K}, \tag{5}$$

K is the total number of links, $K_{\text{satisfied}}$ represents the number of V2V links that satisfy the delay constraint, which can be expressed as:

$$K_{\text{satisfied}} = \sum_{k=1}^{K} 1(T_{\text{left},k} \geq 0 \text{ and } B_{\text{demand},k} \leq 0), \tag{6}$$

where $1(\cdot)$ is an indicator function, which takes the value 1 when the formula is satisfied; otherwise, it is 0. Assuming the maximum tolerable delay is $T_{\max}$, $T_{\text{left},k}$ represents the remaining delay of the k-th V2V link (satisfying $T_{\text{left}} \leq T_{\max}$), and $B_{\text{demand},k}$ represents the remaining amount of data that needs to be transmitted for the current V2V link.

2.3. Formulation of the problem

Overall, the research objective of this paper is to design an optimal joint channel allocation and power control scheme to maximize spectrum resource utilization while striving to meet the communication requirements of different links. This study formulates the resource allocation problem as follows:

$$\max_{\zeta, P} \left(\sum_{m=1}^{M} R_m^c, R_{\text{success}} \right), \tag{7}$$

subject to

$$\sum_{f=1}^{F} \zeta_k^f \leq 1, \tag{8}$$

$$0 \leq p_k^f \leq P_{\max}, \forall k, m, \tag{9}$$

where $\zeta = \left\{ \zeta_1^1, ..., \zeta_k^f, ..., \zeta_K^F \right\}$ and $P = \left\{ p_1^1, ..., p_k^f, ..., p_K^F \right\}$ represent the sub-band selection metrics and the set of power allocations, respectively.

The resource allocation issue expressed in Equation (7) aims to simultaneously maximize the sum data rate for V2I links and the success ratio of transmissions under delay constraints for V2V links. The problem involves multiple objectives in optimization that entail making sequential decisions across multiple transmission time slots within a time constraint, which presents challenges for traditional optimization techniques. To solve this problem, we will give an efficient resource allocation scheme using the MADRL algorithm to solve this problem in the next section.

3. MADRL-based optimization strategy

In the dynamic vehicular communication scenario studied in the present work, we consider each V2V link as an agent, and multiple agents collaborate to explore an unknown vehicular environment and guide the allocation of communication resources to maximize the rewards based on the state changes. In the given section 3.1, we begin by modeling this optimization problem using a Markov decision process (MDP) and define the three essential components: state, action, and reward, respectively. Details about the AMADRL method proposed in this paper are provided in Section 3.2. Finally, Section 3.3 analyzes the algorithm's complexity and communication overhead.

3.1. MDP for resource allocation issues

We first model this MADRL problem as an extended MDP [35]. We define a tuple $(\mathbb{S};\ \mathbb{A};\ \mathbb{R};\ \mathbb{P})$ where $\mathbb{S}$ stands for the state space, $\mathbb{A}$ represents the action space, $\mathbb{R}$ represents the reward, and $\mathbb{P}$ indicates the state transfer probability. At every time slot t, each V2V link acts as a distinct agent, taking action a_t depending on the current state s_t and policy $\pi(a|s)$. Meanwhile, the agents will receive a common reward r_t. The environment will shift to a new state s_{t+1} based on the state transfer probability $p(s', r|s, a)$ (where the state transfer probability and the reward based on the environment's state and the actions performed by the agents, respectively). We outline state, action, and reward separately as detailed below:

1) State space: According to the description in Section 2.1, the state space includes channel information, interference encountered, the count of times a sub-band has been selected by neighbors, load leftover, and time budget leftover. At time slot t, to be precise, the channel information for the k-th V2V link on the f-th sub-band may include the channel gain $g_k^f[t]$ of the V2V link itself, the channel interference $g_{k',k}^f[t]$ from the k'-th V2V transmitter, and the channel interference $\tilde{g}_{m,k}^f[t]$ from the transmitter of the V2I link. We consider that the above information is instantly available at the transmitter of the k-th V2V link via delay-free feedback [36]. In addition, channel interference $g_{k,B}^f[t]$ from the V2V link's transmitter to the BS is introduced into the state. The interference power that the receiver of the k-th V2V link received during the preceding time slot at the f-th sub-band is represented by $I_k^f[t-1]$. We utilize $N_k^f[t-1]$ to denote the count of times sub-band f has been made use of by neighbors in the preceding time slot. Furthermore, the state space contains the residual load $\mathrm{B}_{\text{demand,k}}[\mathrm{t}]$ and the residual time budget $\mathrm{T}_{\text{left,k}}[\mathrm{t}]$ used to satisfy the delay constraints. Thus, at time slot t, the environmental state of the k-th V2V agent could be summarized as:

$$s_t^k = \left\{ \left\{ G_k^f[t] \right\}_{f \in \mathcal{F}}, \left\{ I_k^f[t-1] \right\}_{f \in \mathcal{F}}, \left\{ N_k^f[t-1] \right\}_{f \in \mathcal{F}}, \mathrm{B}_{\text{demand,k}}[\mathrm{t}], \mathrm{T}_{\text{left,k}}[\mathrm{t}] \right\}, \tag{10}$$

where $G_k^f[t] = \left\{ g_k^f[t], g_{k',k}^f[t], \tilde{g}_{m,k}^f[t], g_{k,B}^f[t] \right\}$.

2) Action space: Each agent k performs sub-band selection ζ_k^f and transmission power allocation p_k^f in accordance with the policy π depending on the current state s_t. Therefore, we define the action space as $\mathbb{A} = \{a_1, a_2, ..., a_K\}$, where $a_k = \{\zeta_k^f, p_k^f\}$ denotes the action of each agent k, including discrete binary variables and continuous variables.

3) Reward: RL can represent multiple problems that are difficult to optimize using a flexible reward function. This resource allocation problem aims to select an appropriate transmission sub-band and power for every agent that meets the V2V link delay constraints while reducing interference to the V2I links. In this study, the reward function is composed of three elements: the sum rate of V2I links, the sum rate of V2V links, and the delay condition. The reward function is:

$$r_t^{local} = \left[\lambda \sum_{m=1}^{M} R_m^c + (1-\lambda) \sum_{k=1}^{K} \sum_{f=1}^{F} \zeta_k^f R_k^v \right] - \left(\frac{T_{\max} - \mathrm{T}_{\text{left,k}}[t]}{T_{\max}} \right), \tag{11}$$

where λ denotes the weight, and the quantity $\frac{T_{\max} - \mathrm{T}_{\text{left,k}}[\mathrm{t}]}{T_{\max}}$ is used to denote the penalty, the larger the proportion of the transmission time to the maximum tolerable latency, the larger the penalty.

Unlike the single reward function used in previous RL methods, we define the reward function in Equation (11) as a local reward. On this basis, we also establish a global reward function based on the total interference level experienced by V2V links:

$$r_t^{global} = -\sum_{k=1}^{K} \sum_{f=1}^{F} log_{10}\{\vec{I}_k^f[t]\}, \tag{12}$$

where $\overline{I}_k^f[t] = \sum_{k' \neq k}^{K} \zeta_{k'}^f p_{k'}^f g_{k',k}^f$ represents the interference caused by other V2V links sharing the f-th sub-band. The purpose of using this as a global reward function is to encourage better cooperation among agents, guiding each agent to select resource allocation strategies that cause less interference to other V2V links. Based on the MADRL algorithm, we design a framework with a global and a local critic network to estimate the global and local expected rewards, respectively. This method aims to simultaneously maximize both global and local rewards, addressing the complex optimization problem, which will be detailed in Section 3.2.

3.2. Proposed AMADRL resource allocation method

We improved the MADDPG algorithm and proposed a new resource allocation strategy to solve the complex optimization problem. Meanwhile, we integrated a multi-head AM into the DRL network to better analyze the real-time state information of vehicular communication networks, thereby enhancing resource allocation efficiency.

3.2.1. MADRL network architecture

The DDPG algorithm, based on the actor-critic framework, combines deep learning techniques and is a DRL method capable of handling continuous action space problems. DDPG consists of two main neural networks: the actor network takes the current vehicular network state s_t as input and outputs a specific action $a_t = \pi(s_t)$, where π is the function of the actor network. The critic network estimates the Q-value by taking state-action pairs as input, which is used to evaluate the performance of the actor network. The Q-value function $Q(s_t, a_t)$ represents the cumulative reward R_t obtained by taking actions under policy π. The cumulative discounted reward is defined as:

$$R_t = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \tag{13}$$

where γ is the discount factor, where a value closer to 1 indicates greater emphasis on future rewards. r_t represents the immediate reward. Based on the Bellman equation, this function can be expressed recursively as:

$$Q^\pi(s_t, a_t) = \mathbb{E}_{s_{t+1}} \left[r(s_t, a_t) + \gamma \mathbb{E}_{a_{t+1} \sim \pi} [Q^\pi(s_{t+1}, a_{t+1})] \right]. \tag{14}$$

MADDPG is an extension of DDPG designed for multi-agent scenarios. It estimates the joint Q-value $Q(x, a_1, a_2, \ldots, a_k)$ by taking the information of all agents as input, where x represents the global state that includes the states and observations of all agents, and $a_1, a_2, \ldots, a_k$ represents the actions of all agents. This algorithm effectively mitigates the non-stationarity problem in multi-agent environments.

In this paper, we improved the above algorithm by training a global critic network and multiple local critic networks to jointly calculate the Q-value. The global critic network is shared among all agents, similar to the MADDPG algorithm. It maintains the global reward by inputting information from all agents, promoting cooperation among agents and improving overall performance. The local critic networks are similar to the standard single-agent DDPG, taking only the local observation state and action of a specific agent as input to maintain the local reward, balancing individual constraints and objectives. The local critic networks reduce the impact of uncertainty from global information on individual policies, thereby enhancing training stability. By adopting a framework similar to centralized training and decentralized execution, we update each agent's policy using feedback from both the global and local critics in a multi-agent environment, aiming to maximize both global and local rewards. The specific framework of the algorithm is shown in Fig. 2.

At each time step, agents observe states, select actions, calculate global and local rewards, and store the data in a replay buffer. The algorithm then updates the global and local critic networks by sampling data from the buffer. Finally, the actor network is updated every two episodes based on the global and local Q-values. By integrating global and local rewards, this method optimizes resource allocation in vehicular communication networks and reduces computational costs and decision-making complexity.

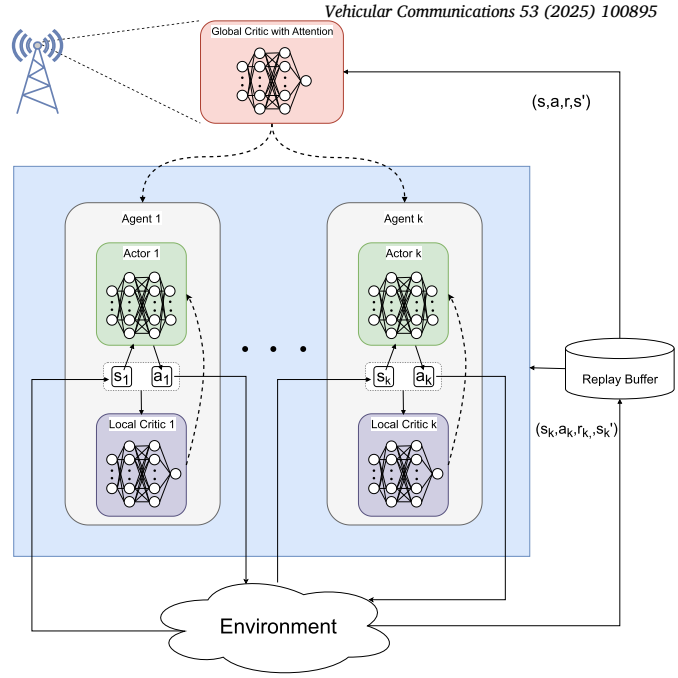

Fig. 2. AMADRL algorithm framework diagram.

Specifically, assuming a vehicular network environment with K agents, the set of policy networks is denoted by $\pi = \{\pi_1, \pi_2, ..., \pi_K\}$, with the actor network parameters represented by $\theta = \{\theta_1, \theta_2, ..., \theta_K\}$. The local Q-function of the k-th agent is denoted as $Q_{\mu_k}^k$, with parameters $\mu = \{\mu_1, \mu_2, \ldots, \mu_k\}$, while the global Q-function is denoted as Q_φ^{global}, parameterized by φ. Based on single-agent DDPG and MADDPG, we improve the algorithm, and the improved policy gradient of each agent k is updated based on the global and the local Q-value, expressed as:

$$\nabla J(\theta_k) = \mathbb{E}_{s,a\sim B} \left[\nabla_{\theta_k} \pi_k(a_k|o_k) \nabla_{a_k} Q_\varphi^{global}(s,a) \right] + \mathbb{E}_{o_k,a_k\sim B} \left[\nabla_{\theta_k} \pi_k(a_k|o_k) \nabla_{a_k} Q_{\mu_k}^k(o_k,a_k) \right] \tag{15}$$

where o_k represents the observation received by agent k, $\mathbf{s} = (o_1, o_2, \ldots, o_k)$ and $\mathbf{a} = (a_1, a_2, \ldots, a_k)$ represent the state and action information of all agents. $a_k = \pi_k(o_k) + \varepsilon$ is the action selected by agent k based on policy π and observation o_k. To improve exploration during action selection, we add random noise $\varepsilon \sim \text{clip}(\mathcal{N}(0, 0.2), -0.5, 0.5)$. B denotes the experience replay buffer. The actor network optimizes the policy by maximizing the Q-value outputs of both the global and local critic networks.

The global critic network is updated by minimizing the error with the target value:

$$\mathcal{L}(\varphi) = \mathbb{E}_{s,a,r,s'} \left[\left(Q_\varphi^{global}(\mathbf{s}, \mathbf{a}) - y_{global} \right)^2 \right], \tag{16}$$

where the target value function y_{global} is:

$$y_{global} = r_{global} + \gamma Q_{\varphi'}^{global}(s', a')|_{a'_k = \pi'_k(o'_k) + \varepsilon}. \tag{17}$$

Here, the set of target policy networks is denoted by $\pi' = (\pi'_1, \pi'_2, \ldots, \pi'_k)$, and its network parameters are $\theta' = (\theta'_1, \theta'_2, \ldots, \theta'_k)$.

Similarly, the local critic network $Q_{\mu_k}^k$ is updated as follows:

$$\mathcal{L}(\mu_k) = \mathbb{E}_{o_k,a_k,r_k,o'_k} \left[\left(Q_{\mu_k}^k(\mathbf{o_k}, \mathbf{a_k}) - y_{local} \right)^2 \right], \tag{18}$$

where

$$y_{local} = r^k_{local} + \gamma Q^k_{\mu'_k}(\boldsymbol{o}'_k, \boldsymbol{a}'_k)|_{\boldsymbol{a}'_k = \boldsymbol{\pi}'_k(o'_k)+\varepsilon}. \tag{19}$$

During the execution phase, the critic network does not participate in decision-making. Each agent k independently selects actions using its actor network based on local observations o_k.

The pseudocode described in Algorithm 1 outlines the specific steps of the algorithm. Through the aforementioned design, the improved AMADRL algorithm not only meets the transmission requirements of V2V and V2I links but also promotes cooperation among agents, enhancing the efficiency of resource allocation and overall performance.

Algorithm 1: The main steps of the AMADRL method.

Initialize environment parameters
Initialize actor and critic networks for each agent
Initialize global critic network Q^{global}_{φ}
Initialize each target networks
Initialize experience replay buffer
foreach *episode in total_episodes* **do**
  if *episode is at specific interval* **then**
    Reinitialize the environment
  end if
  foreach *time slot t* **do**
    foreach *agent in agents* **do**
      Get current agent's observe o^t_k
      Select action based on current state $a^t_k = \pi_{\theta_k}(o^t_k)$
    end foreach
    Execute action $\mathbf{a}^t = [a^t_1, \ldots, a^t_K]$ and state $\mathbf{s}^t = [o^t_1, \ldots, o^t_K]$
    Obtain reward r^{global}_t and r^{local}_t
    Store $(\mathbf{s}^t, \mathbf{a}^t, \mathbf{r}^{local}_t, r^{global}_t, \mathbf{s}^{t+1})$ in buffer B
  end foreach
  if *replay buffer is sufficiently large* **then**
    Randomly select a small batch of data S from the replay buffer
  end if
  Set $y_{global} = r_{global} + \gamma Q^{global}_{\varphi'}(s', a')|_{a'_k = \pi'_k(o'_k)+\varepsilon}$
  Update global critic by minimizing $\frac{1}{S}\sum\left(Q^{global}_{\varphi}(\mathrm{s}, \mathrm{a}) - y_{global}\right)^2$
  Update target network: $\varphi' \leftarrow \tau\varphi + (1-\tau)\varphi'$
  if *episode mod d* **then**
    foreach *agent k* **do**
      Set $y_{local} = r^k_{local} + \gamma Q^k_{\mu'_k}(o'_k, a'_k)|_{a'_k = \pi'_k(o'_k)+\varepsilon}$
      Update local critic by minimizing
      $\frac{1}{S}\sum\left(Q^k_{\mu_k}(\mathbf{o_k}, \mathbf{a_k}) - y_{local}\right)^2$
      Update actor networks :
      $$\nabla J(\theta_k) = \frac{1}{S}\left[\sum\left[\nabla_{\theta_k}\pi_k(\mathrm{a_k}|\mathrm{o_k})\nabla_{\mathrm{a_k}} Q^{\mathrm{global}}_{\varphi}(\mathrm{s}, \mathrm{a})\right] + \sum\left[\nabla_{\theta_k}\pi_k(\mathrm{a_k}|\mathrm{o_k})\nabla_{\mathrm{a_k}} Q^k_{\mu_k}(\mathrm{o_k}, \mathrm{a_k})\right]\right]$$
    end foreach
    Update target networks : $\mu' \leftarrow \tau\mu + (1-\tau)\mu'$ and $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$
  end if
end foreach

3.2.2. Application of attention mechanisms in resource allocation

Since DRL algorithms typically use multiple linear layers to construct the critic network, they struggle to capture the key features in the input information. Therefore, in the global critic network, we incorporated a multi-head AM to enhance feature representation capabilities and resource allocation efficiency. The AM is introduced between the first and second linear layers to weight the extracted features, enabling the model to better focus on the parts most relevant to the current policy decisions and quickly adjust resource allocation strategies. The specific network structure is shown in Fig. 3.

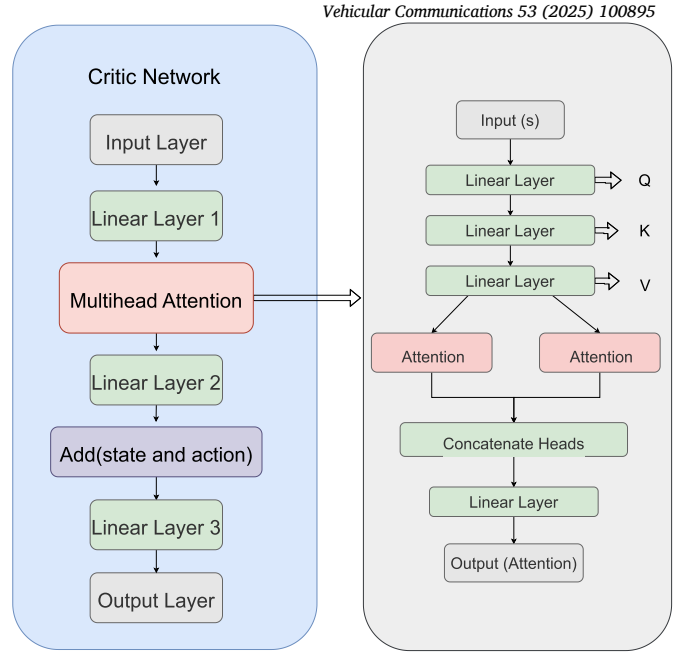

Fig. 3. The structure of the global critic network with an added attention layer.

The core idea of the AM is to enable the model to focus on key parts by computing the importance weights of different input elements. The AM typically has three inputs: the query vector (Q), the key vector (K), and the value vector (V). The multi-head attention mechanism, as an extension of the standard AM, learns in parallel by mapping input data into different subspaces. In this paper, we set the number of heads in the multi-head attention mechanism to 2. The vehicle network state s is transformed through three linear layers, mapping it to Q_n, K_n, and V_n $(n = 1, 2)$, which are used as the three inputs for the n-th attention head. Each attention head computes dot-product attention on the three vectors Q_n, K_n, and V_n, where the output of the n-th attention head is:

$$\mathrm{Attention}(Q_n, K_n, V_n) = \mathrm{softmax}\left(\frac{Q_n K_n^T}{\sqrt{\mathrm{d_k}}}\right) V_n, \tag{20}$$

where $\mathrm{d_k}$ is the dimension of the key vector, serving as a scaling factor to mitigate the potential issue of excessively large values resulting from vector dot products, thereby avoiding gradient instability. In the formula, the weight ω_n is first calculated by performing the dot product between vectors Q_n and K_n. By multiplying ω_n with V_n, the weight is assigned to different state features, enabling the selection of important information about the vehicle network state. The formula for ω_n is as follows:

$$\omega_n = softmax\left(\frac{Q_n K_n^T}{\sqrt{\mathrm{d_k}}}\right). \tag{21}$$

By calculating attention independently for each head, concatenating the results of all attention heads, and processing them through a linear layer transformation, the final multi-head attention output is obtained.

3.3. Computational complexity and communication overhead

For the AMADRL algorithm proposed in this paper, we assume that $\mathrm{N}^{\mathrm{a}}_{\mathrm{l}}$ and $\mathrm{N}^{\mathrm{c}}_{\mathrm{l}}$ represent the number of neurons in the l-th layer of the actor network and critic network, respectively.

1) Global critic network: Since the AM is introduced between the first linear layer and the second linear layer, its computational complexity primarily comes from calculating the values of Q, K, V, and the atten-

Table 2
Simulation Parameters.

Definitions	Notion	Value
Bandwidth	B	1.5 MHz
Carrier frequency	C	2 GHz
Vehicle speed	μ	36 km/h
Latency constraints for V2V links	$T_{\max}$	100 ms
Noise power	σ^2	-114 dBi
Discount factor	γ	0.99
V2I transmit power level	p_m	23 dBm
V2V transmit power level	p_k^f	[0,23] dBm
Actor-network learning rate	α	0.0001
Critic-network learning rate	β	0.001
Soft update parameters	τ	[0,1]
Weight	λ	0.7

tion weights. Therefore, the complexity is $\mathcal{O}(\mathrm{N}_l^{c^2})$, while the complexity of the remaining layers is $\mathcal{O}\left(\sum_{l=2}^{l=\mathrm{H_c}-1}(\mathrm{N}_{l-1}^{c}\mathrm{N}_l^{c}+\mathrm{N}_l^{c}\mathrm{N}_{l+1}^{c})\right)$, where $\mathrm{H_c}$ is the number of layers in the critic network. Thus, the total complexity of the global critic network is $\mathcal{O}\left(\left(\sum_{l=2}^{l=\mathrm{H_c}-1}(\mathrm{N}_{l-1}^{c}\mathrm{N}_l^{c}+\mathrm{N}_l^{c}\mathrm{N}_{l+1}^{c})+\mathrm{N}_l^{c\,2}\right)\cdot \mathrm{M}\cdot\mathrm{E}\cdot\mathrm{S}\right)$, where M is the mini-batch size, E is the number of training episodes, and S is the number of training steps per episode.

2) Local critic network: Since the local critic network does not incorporate the AM, its total complexity is $\mathcal{O}\left(\left(\sum_{l=2}^{l=\mathrm{H_c}-1}\left(\mathrm{N}_{l-1}^{c}\mathrm{N}_l^{c}+\mathrm{N}_l^{c}\mathrm{N}_{l+1}^{c}\right)\right)\cdot \mathrm{M}\cdot\mathrm{E}\cdot\mathrm{S}\cdot\mathrm{K}\right)$, where K is the number of agents.

3) Actor network: Similar to the aforementioned studies, the total complexity of the actor network is $O\left(\left(\sum_{l=2}^{l=\mathrm{H_a}-1}\left(\mathrm{N}_{l-1}^{a}\mathrm{N}_l^{a}+\mathrm{N}_l^{a}\mathrm{N}_{l+1}^{a}\right)\right)\cdot \mathrm{M}\cdot\mathrm{E}\cdot\mathrm{S}\cdot\mathrm{K}\right)$, where $\mathrm{H_a}$ is the number of layers in the actor network.

Communication overhead is crucial for evaluating the effectiveness of the proposed resource allocation strategy. Excessive communication overhead may significantly reduce the framework's efficiency. We focus on the number of interactions between agents, RSUs, and other agents during the training process. In our proposed algorithm, the overhead can be reduced to K (where K is the number of agents), thus reducing the communication burden.

4. Simulation results

In this section, we present the experimental results and evaluations to demonstrate the performance of the proposed AMADRL algorithm.

4.1. Simulation environment

We simulated the vehicular network environment using the Manhattan case defined in 3GPP TR 36.885 [37]. Multiple vehicles are randomly distributed onto the road according to a spatial Poisson process. M V2I links are formed by M vehicles, respectively, each of which is scheduled to communicate with three surrounding vehicles to form K V2V links. All simulation results were generated using Tensorflow 2.6.0 on the Python 3.8 environment. The critic network of each V2V agent consists of multiple layers, including three fully connected layers (containing 1024, 512, and 256 neurons, respectively) and the global critic network also includes a multi-head attention layer. The actor network comprises two fully connected layers, containing 1024 and 512 neurons, respectively. We set the delay update factor $d = 2$ for the actor network to reduce instability during the training process. We set the weight of the local reward function to 0.7, indicating that this study places greater emphasis on reducing interference to the V2I link. Our activation function is the rectified linear unit (ReLU), defined as $f(x) = \max(0, x)$, and the parameters are updated by iterative training using the adaptive moment estimation (Adam) [38]. Table 2 lists the main parameters.

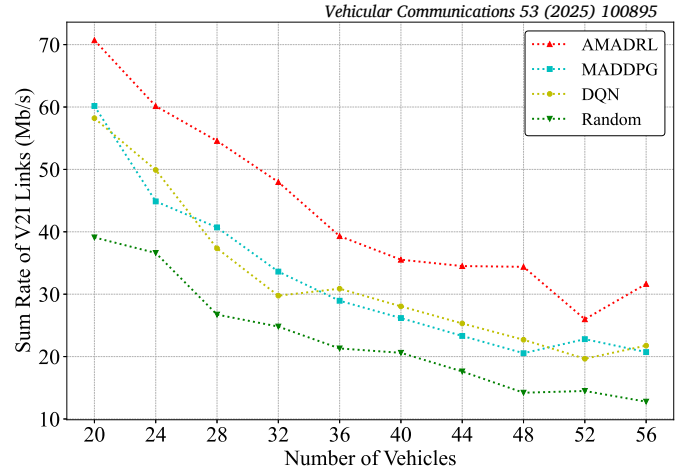

Fig. 4. The rate of V2I links versus the number of vehicles.

4.2. Baseline methods

We assessed the proposed AMADRL strategy in comparison to the three approaches outlined below:

1) Random scheme. At every time step, the agent selects a spectrum sub-band and power at random.

2) A resource allocation scheme based on the DQN algorithm, in which continuous power is discretized into three power levels: [23, 10, 5] dBm. The algorithm adopts a dual-network structure consisting of a Q-network and a target network.

3) The MADDPG algorithm, based on a centralized training and decentralized execution framework, consists of actor and critic networks and is capable of efficiently learning agents' policies in a continuous action space.

4.3. Experimental results

As in Fig. 4, we established the number of vehicles as [20,24,28,32, 36,40,44,48,52,56] to observe the correlation between the V2I links' sum rate and the vehicle count under different resource allocation methods. We can observe that with the increase in vehicles, the V2I links' sum rate falls continuously, leading to a reduction in the performance of all methods. This is because the number of V2V links increases as vehicle counts increase. The limited spectrum will be allocated to more V2V links, leading to greater interference with V2I links, causing a continuous reduction of the V2I link transmission capacity. However, all three DRL-based resource allocation methods outperform the random solution, demonstrating the effectiveness of DRL algorithms in resource allocation for vehicular networks. Compared to the DQN and MADDPG algorithms, the method proposed in this paper achieves better gains in the sum rate of V2I links. This is because the DQN algorithm quantizes continuous transmission power, leading to a reduction in the precision of transmission power and thus decreasing the adaptability of the policy to channel states. In contrast, the AMADRL algorithm employs a dual-critic network structure, which not only achieves global optimization but also meets individual requirements. This allows multiple agents to better balance cooperation and competition, enabling more rational strategy choices. Additionally, the integration of the AM prioritizes processing information that is most valuable for reducing interference, enabling rapid adjustments to resource allocation strategies and effectively reducing interference with V2I links.

Fig. 5 illustrates the relationship between the proportion of V2V links successfully meeting delay constraints and the number of vehicles under different resource allocation methods. It can also be observed that the success rate of V2V link transmissions decreases as the number of vehicles increases. This is because vehicles need to share a fixed spectrum resource, and as the count of V2V links grows under limited wireless

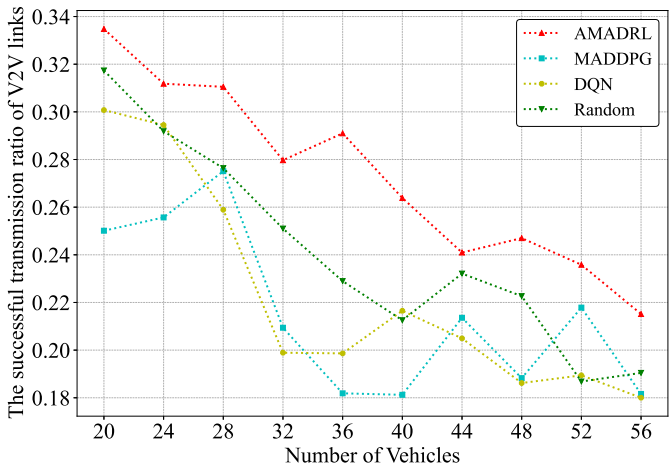

Fig. 5. The successful transmission ratio of V2V links versus the number vehicles.

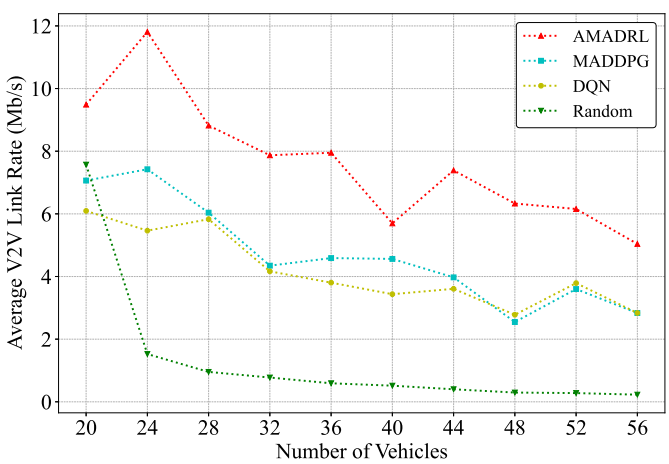

Fig. 6. The average rate of V2V links with an increasing number of vehicles under different algorithms.

resources, the bandwidth allocated to each link decreases, leading to reduced transmission speeds. Consequently, none of the methods can meet the required communication demands, and the proportion of vehicles able to satisfy the delay constraints steadily declines. However, the method proposed in this paper demonstrates superior performance compared to other baseline methods. For instance, when the number of vehicles reaches 56, the proposed algorithm achieves a transmission success rate of 21.5%, compared to 18.2%, 18%, and 19.1% for MADDPG, DQN, and Random methods, respectively. This is because the proposed algorithm not only enables each agent to optimize its own actions but also facilitates collaboration to avoid conflicts with other vehicles. By dynamically adjusting sub-band and power allocation, the proposed AMADRL method allocates more resources to links that are at risk of violating delay constraints, effectively mitigating the proportion of transmission failures.

Fig. 6 shows the average rate of V2V links under different resource allocation methods as the number of vehicles increases. As can be seen from the figure, the average rate of V2V links decreases with the increase in the number of vehicles. This is because, as the number of vehicles grows, more V2V links communicate simultaneously, resulting in fewer resources allocated to each V2V link and increased interference between links. However, compared to other methods, the proposed approach demonstrates better performance due to the constraints in the reward function, which reduce the interference caused by V2V agents to other V2V links.

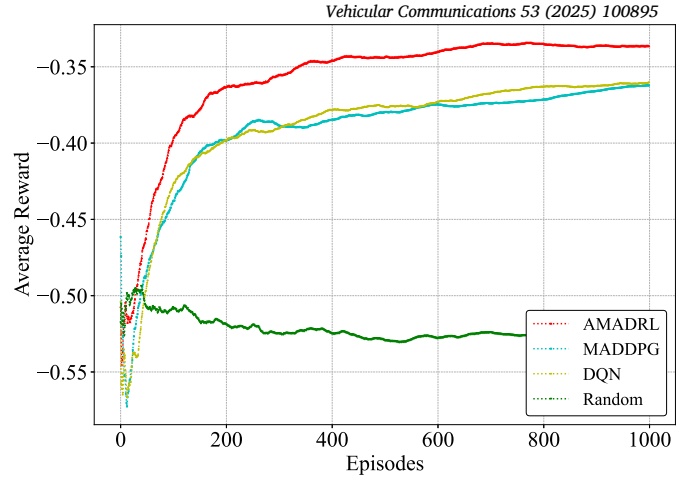

Fig. 7. Reward for each training episode.

Fig. 7 shows the average cumulative rewards with the increase in training iterations, analyzing the convergence performance of the proposed AMADRL method. It can be observed that the rewards obtained by DRL-based resource allocation methods are all greater than those of the random method, which only performs random policy searches based on immediate rewards. With the increase in training iterations, the cumulative rewards of DRL-based methods improve and gradually stabilize, demonstrating the effectiveness of the DRL algorithms. Due to the exploration strategies, the curves may exhibit oscillations. After 650 episodes, the cumulative rewards of the proposed algorithm begin to stabilize. With an increasing number of V2V links, the dimensionality of the state space grows accordingly. As a result, the DQN algorithm requires a larger experience replay buffer and a more complex network structure to approximate the Q-value function, significantly increasing the training difficulty. For the MADDPG algorithm, the input to the critic network includes information from all agents. This shared critic network for all agents makes the learning task more complex, often leading to unstable training and slower convergence. In contrast, the proposed method separates global and local critic tasks, allowing the global critic network to focus on the overall network dynamics while the local critic network targets the individual goals of agents. Additionally, the inclusion of the attention mechanism effectively improves training efficiency. Compared to other methods, the proposed approach achieves the best reward performance with the fastest convergence speed and the most stable learning process (less fluctuation), further validating the effectiveness of the model.

Fig. 8 illustrates the average transmission power selection of V2V links under different algorithms as the number of vehicles increases. It can be observed that, compared to other methods, the proposed approach tends to select lower power levels. This is because lower power levels are often the result of the optimization algorithm aiming to minimize unnecessary energy consumption while meeting communication quality requirements, thereby achieving efficient resource utilization and reducing interference. As the number of vehicles increases, communication demands become more intensive, and communication interference becomes more severe. To overcome channel fading and interference, vehicles may need to increase their transmission power to ensure communication reliability, leading to a rise in power usage. However, this trend is constrained by interference management mechanisms. In this study, the global reward function is designed to limit the interference caused by agents to others. Therefore, as the number of vehicles grows, the transmission power levels show a decreasing trend to mitigate interference while maintaining a certain level of communication quality.

The Table 3 records the average computation time required by different algorithms. Since the network structure of the DQN algorithm is lighter compared to MADDPG and the proposed algorithm, it has the

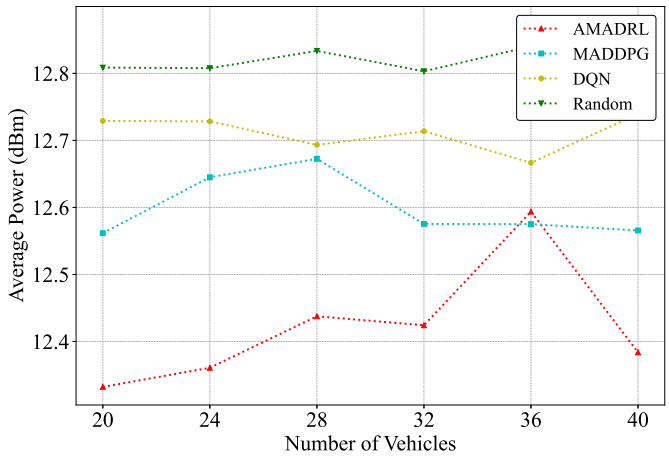

Fig. 8. Power selection trends under different algorithms as the number of vehicles increases.

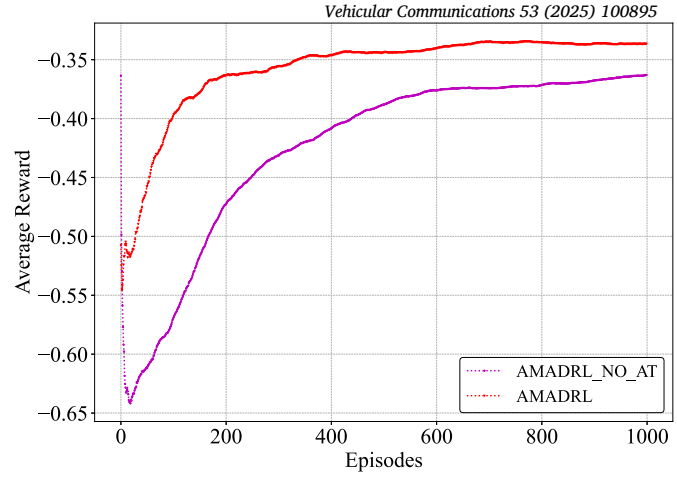

Fig. 9. Reward for each training episode.

Table 3
The average computation time required by different algorithms.

Algorithms	Average computation time (s)
Random	0.0543
DQN	0.4642
MADDPG	1.0524
Proposed algorithm	1.0344

shortest execution time. The proposed algorithm adopts a dual-critic network structure, improving upon the traditional MADDPG algorithm by avoiding the problem of using a single critic network to handle the states and actions of all agents, which results in complexity growing linearly with the number of agents. Although the introduction of the attention mechanism increases computation time, the total computation time is still less than that of traditional MADDPG, demonstrating the superiority of the proposed algorithm.

To gain a deeper understanding of the impact of introducing the attention mechanism on performance, we removed the attention mechanism module, eliminating the need to process input data such as states through the attention layer, and compared this with the improved algorithm. The results are shown in the Fig. 9. It can be observed that the introduction of the attention mechanism allows the critic network to dynamically focus on key features, significantly reducing redundant computations and improving training efficiency. The actor network is able to learn more efficient resource allocation strategies, enabling the model to converge faster and reduce instability in sparse reward environments.

5. Conclusion

This paper investigates the problem of wireless communication resource allocation in dynamic vehicular communication networks. We improved the traditional MADRL framework and proposed a novel resource allocation method, AMADRL. This method employs global and a local critic network to estimate corresponding global and local reward functions, achieving joint optimization of spectrum allocation and power control in vehicular networks while meeting the communication requirements of different links. To enhance the model's ability to capture important features, we integrated an attention mechanism into the algorithm, enabling the framework to selectively focus on critical information and improve task efficiency. Experimental results validate that the proposed algorithm effectively satisfies the transmission requirements of different links and demonstrates the advantages of incorporating the attention mechanism into the algorithm. For future work, extending the proposed method to address resource allocation in UAV-assisted vehicular network communications presents an intriguing direction.

CRediT authorship contribution statement

Zhibin Liu: Writing – review & editing, Writing – original draft, Software, Resources, Formal analysis. **Yifei Deng:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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